분석 시각화

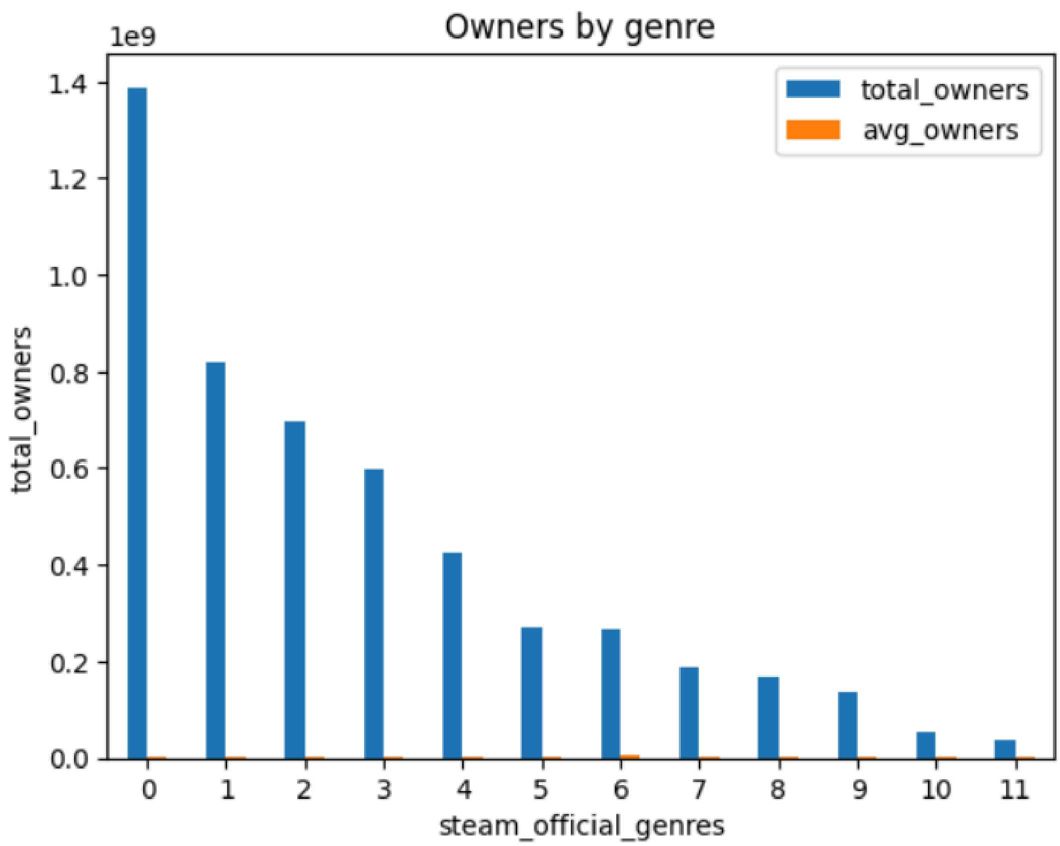
```
df=pd.read_csv("C:/Mydisk/workspaces/SQL_project/genre_user.csv")
df.head()
```

	steam_official_genres	total_owners	avg_owners
0	Action	1388608890	2.760654e+06
1	Adventure	817216260	2.190928e+06
2	Indie	695782950	2.089438e+06
3	RPG	597676890	2.157678e+06
4	Simulation	425611110	1.514630e+06

장르, 총 유저 수, 평균 유저 수만 기록 된 CSV 데이터프레임을 이용.

장르 별 총 유저 수(막대)
x축: 장르, y축: 유저수

```python
df.plot.bar(xlabel='steam_official_genres',
            ylabel='total_owners',
            rot=0,
            title='Owners by genre')
plt.show()
```



장르 이름 출력 안됨. 평균 유저수가 함께 출력 됨.

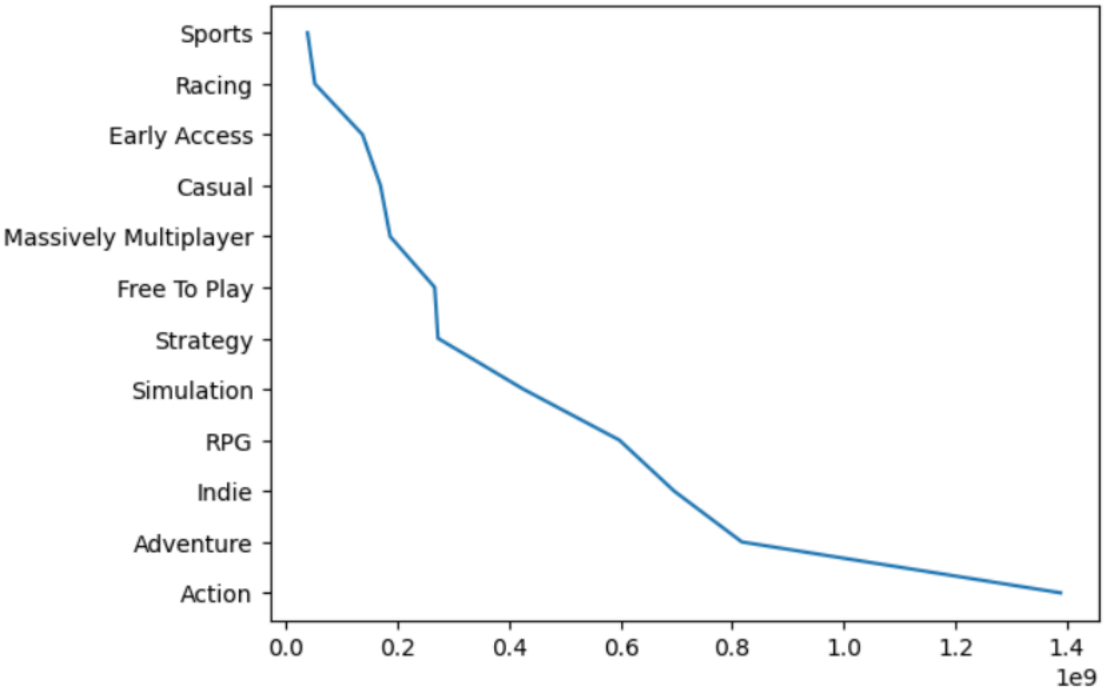
```
column=['steam_official_genres', 'total_owners']
owners_by_genre=df[column].copy()
owners_by_genre
```

	steam_official_genres	total_owners
0	Action	1388608890
1	Adventure	817216260
2	Indie	695782950
3	RPG	597676890
4	Simulation	425611110
5	Strategy	272506170
6	Free To Play	266590830
7	Massively Multiplayer	186719040
8	Casual	169345560
9	Early Access	137200080
10	Racing	51681870
11	Sports	38629380

필요 컬럼만 분리.

```
plt.plot(owners_by_genre['total_owners'], owners_by_genre['steam_official_genres'])
```

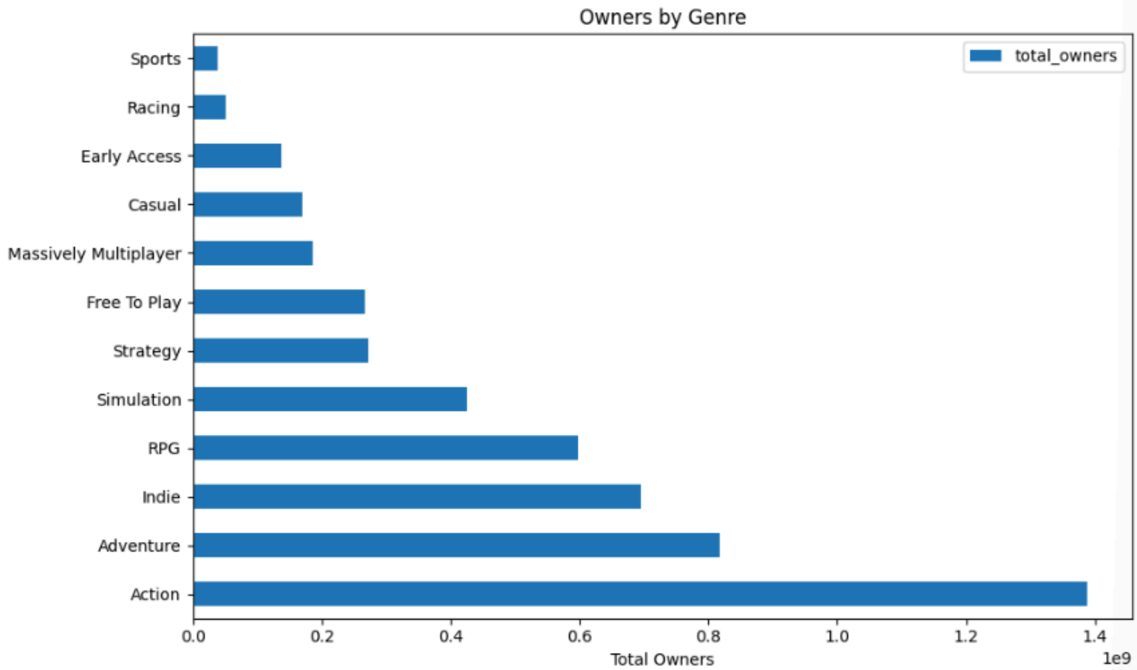
```
[<matplotlib.lines.Line2D at 0x24266267550>]
```



선 그래프 출력. 하지만 원하는 것은 막대그래프.

```
owners_by_genre.set_index('steam_official_genres')[['total_owners']].plot.barh(
    figsize=(10, 6), title='Owners by Genre', xlabel='Total Owners', ylabel=''
)

plt.tight_layout()
plt.show()
```



set_index를 이용하여 y축에 장르 명을 출력. 막대그래프를 채택하여 장르별 유저 분포도를 한 눈에 볼 수 있도록 설정.

장르 별 평균 유저 수 및 긍정평가 비율(산점도)
x축: 유저수, y축: 긍정평가 비율

```python
df2=pd.read_csv("C:/Mydisk/workspaces/SQL_project/genre_avg.csv")
df2
```

	steam_official_genres	total_owners	avg_owners	avg_positive
0	Action	1388608890	2.760654e+06	82.3141
1	Adventure	817216260	2.190928e+06	83.5282
2	Indie	695782950	2.089438e+06	87.1201
3	RPG	597676890	2.157678e+06	81.1877
4	Simulation	425611110	1.514630e+06	84.6833
5	Strategy	272506170	1.449501e+06	83.4787
6	Free To Play	266590830	5.126747e+06	69.5962
7	Massively Multiplayer	186719040	2.917485e+06	71.7969
8	Casual	169345560	1.282921e+06	86.6970
9	Early Access	137200080	1.193044e+06	82.8696
10	Racing	51681870	1.292047e+06	81.9250
11	Sports	38629380	9.197471e+05	78.3810

총 유저수, 평균 유저수, 평균 긍정 평가 비율이 포함된 데이터프레임 생성.

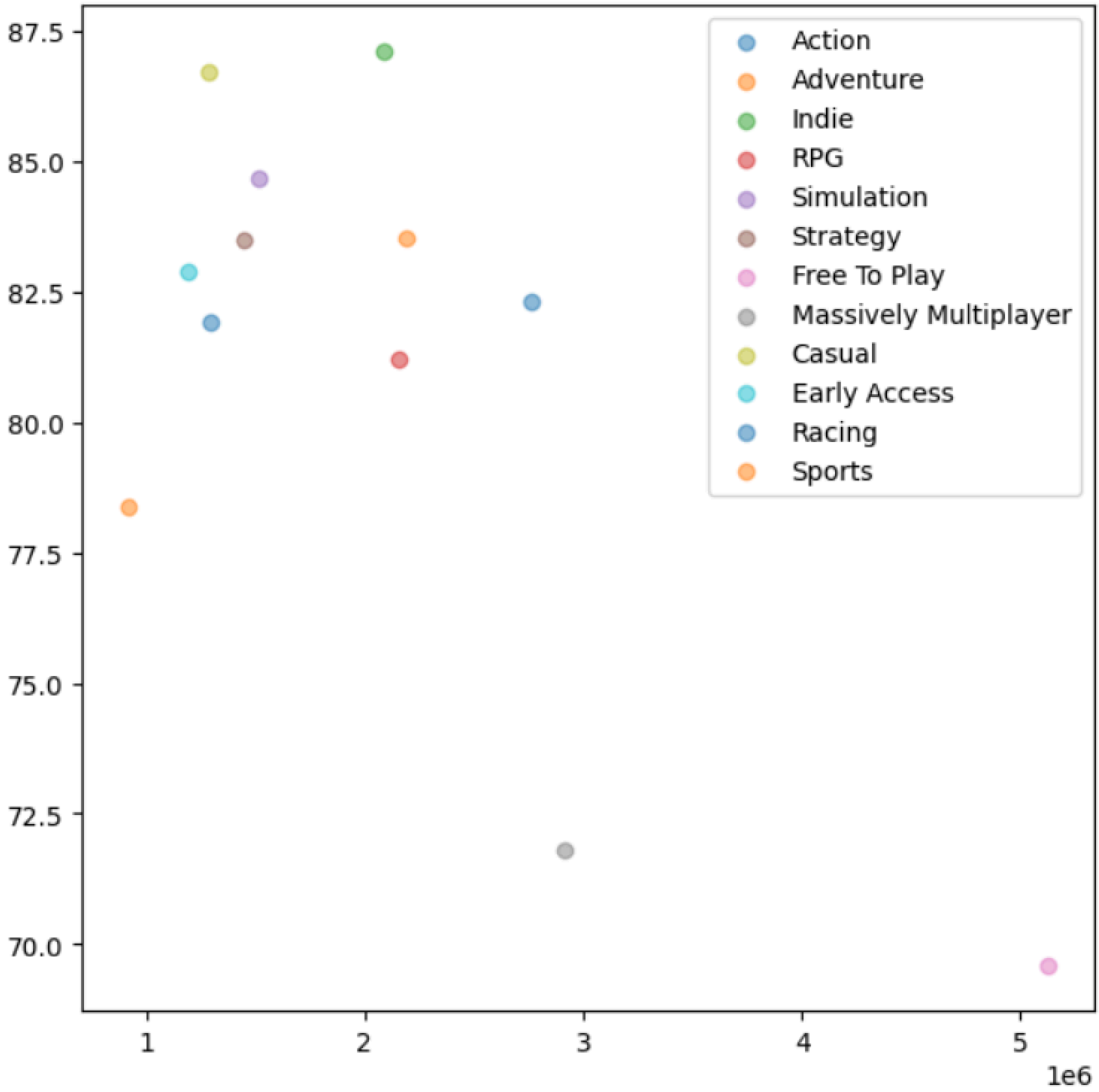
```
column=['steam_official_genres', 'avg_owners', 'avg_positive']
genre_avg=df2[column].copy()
genre_avg
```

	steam_official_genres	avg_owners	avg_positive
0	Action	2.760654e+06	82.3141
1	Adventure	2.190928e+06	83.5282
2	Indie	2.089438e+06	87.1201
3	RPG	2.157678e+06	81.1877
4	Simulation	1.514630e+06	84.6833
5	Strategy	1.449501e+06	83.4787
6	Free To Play	5.126747e+06	69.5962
7	Massively Multiplayer	2.917485e+06	71.7969
8	Casual	1.282921e+06	86.6970
9	Early Access	1.193044e+06	82.8696
10	Racing	1.292047e+06	81.9250
11	Sports	9.197471e+05	78.3810

필요 컬럼 추출.

```python
fig=plt.figure(figsize=(7,7))
ax=fig.add_subplot(111)

for genres in genre_avg['steam_official_genres'].unique():
    sub=genre_avg[genre_avg['steam_official_genres']==genres]
    ax.scatter(x=sub['avg_owners'],
               y=sub['avg_positive'],
               alpha=0.5,
               label=genres)
ax.legend()
plt.show()
```



산점도 출력.